NSBM Green University

Faculty of Computing

CN204.3 -Network Security

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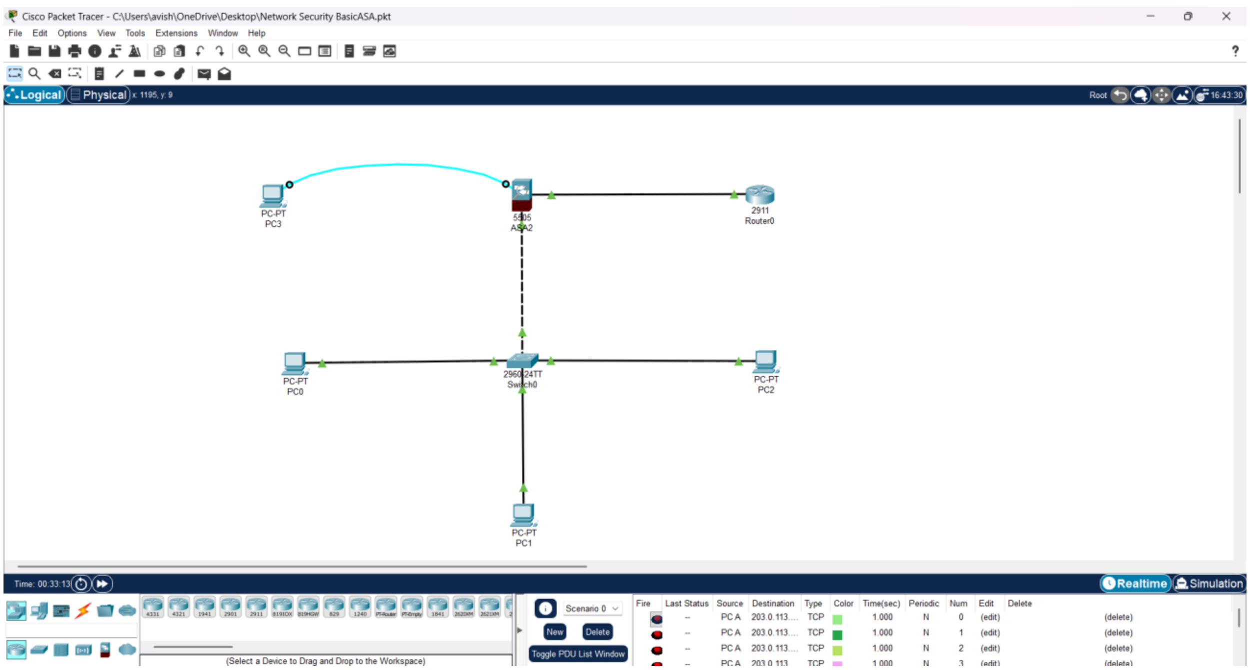
Course Work

Group R

1. Introduction

➢ This project focuses on the configuration of a Cisco ASA firewall to secure an internal LAN from an external router, establishing a controlled security boundary. The aim was to validate basic connectivity and then enforce access control through ACLs. Initially, ICMP traffic was tested to confirm routing and communication. Once verified, ACLs were configured to allow only HTTP traffic while blocking all other protocols

2. Network Topology



➢ **The network consisted of:**

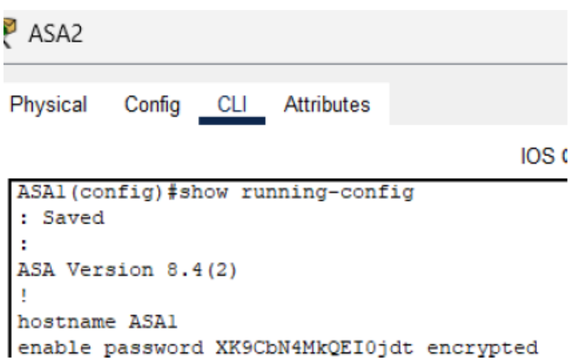
- An ASA firewall configured with inside and outside VLAN interfaces.
- An internal LAN with PCs configured in the inside network.
- An external router simulating Internet access.

3. Interface and Routing Configuration

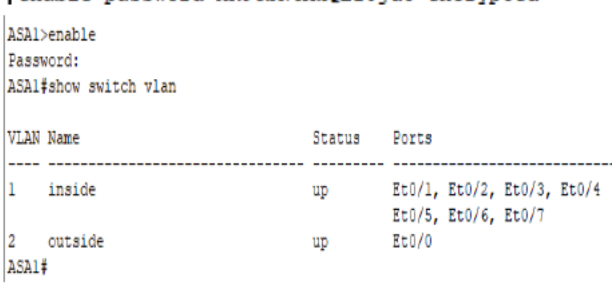
ASA Interface Verification

Commands used:

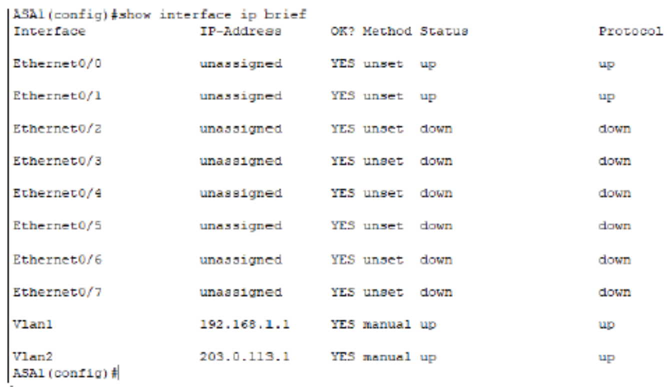
```
show interface ip brief
show switch vlan
show route
```



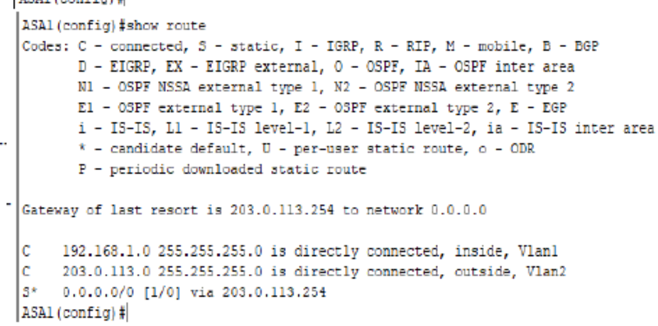
```
ASA2
Physical   Config   CLI   Attributes
IOS (
ASA1(config)#show running-config
: Saved
:
ASA Version 8.4(2)
!
hostname ASA1
enable password XK9CbN4MkQEI0jdt encrypted
```



```
ASA1>enable
Password:
ASA1#show switch vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- ----------------------------
1    inside                           up        Et0/1, Et0/2, Et0/3, Et0/4
                                                Et0/5, Et0/6, Et0/7
2    outside                          up        Et0/0
ASA1#
```



```
ASA1(config)#show interface ip brief
Interface              IP-Address      OK? Method Status                Protocol

Ethernet0/0            unassigned      YES unset  up                    up

Ethernet0/1            unassigned      YES unset  up                    up

Ethernet0/2            unassigned      YES unset  down                  down

Ethernet0/3            unassigned      YES unset  down                  down

Ethernet0/4            unassigned      YES unset  down                  down

Ethernet0/5            unassigned      YES unset  down                  down

Ethernet0/6            unassigned      YES unset  down                  down

Ethernet0/7            unassigned      YES unset  down                  down

Vlan1                  192.168.1.1     YES manual up                    up

Vlan2                  203.0.113.1     YES manual up                    up
ASA1(config)#
```



```
ASA1(config)#show route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.254 to network 0.0.0.0

C    192.168.1.0 255.255.255.0 is directly connected, inside, Vlan1
C    203.0.113.0 255.255.255.0 is directly connected, outside, Vlan2
S*   0.0.0.0/0 [1/0] via 203.0.113.254
ASA1(config)#
```

- Results confirmed that inside and outside interfaces were active, VLANs were properly assigned, and a static default route was configured.

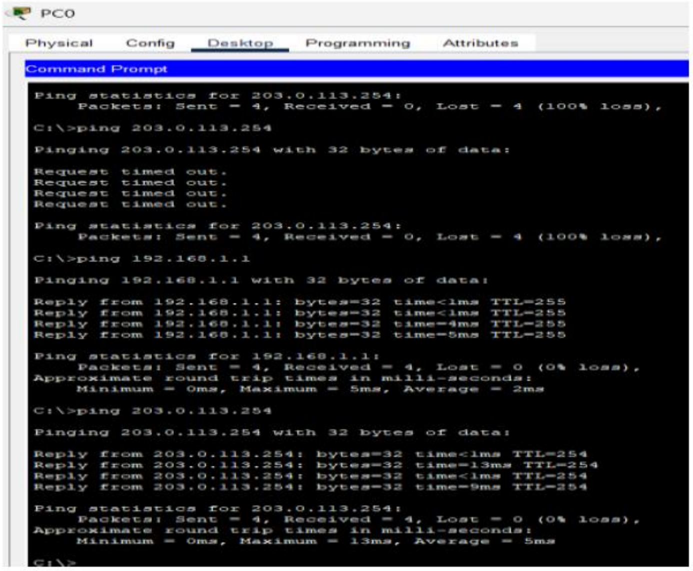
Router Configuration Verification

Commands executed:

```
show ip interface brief
show ip route
```

Output verified Router0 had IP 203.0.113.254, interfaces were up, and a static route to the internal LAN was present.

4. Baseline Connectivity (Before ACLs)




```
Router0#ping 203.0.113.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/11 ms

Router0#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router0#
```

- ➢ Baseline ping tests confirmed that devices could communicate with each other before ACL restrictions were applied. This validated correct IP addressing and routing setup.

5. ACL Testing with ICMP (Temporary)

```
Router0#ping 203.0.113.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/7 ms

Router0#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/13 ms

Router0#
```

- A temporary ACL was placed on the outside interface to permit ICMP traffic for connectivity testing:

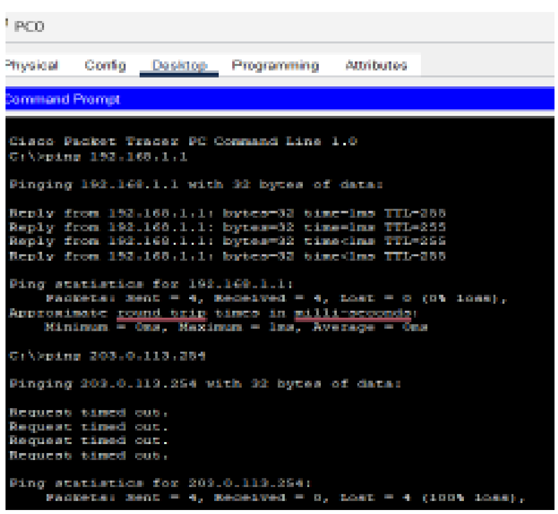
  access-list OUTSIDE-IN extended permit icmp host 203.0.113.254 192.168.1.0 255.255.255.0
  access-group OUTSIDE-IN in interface outside

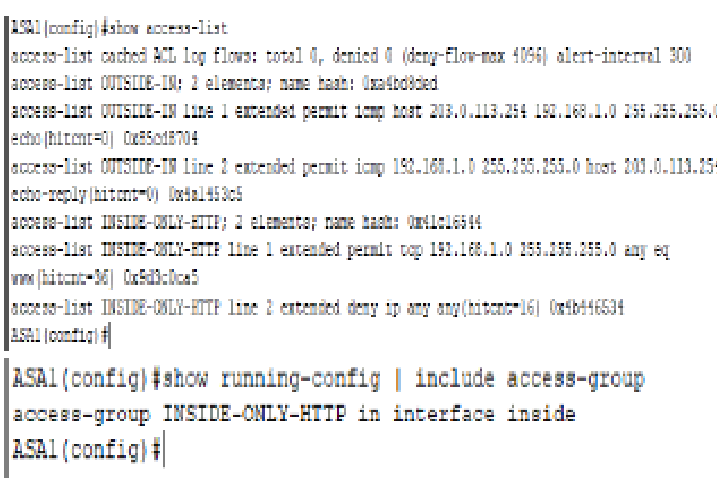
  This allowed pings to verify correct connectivity before moving to stricter ACLs.

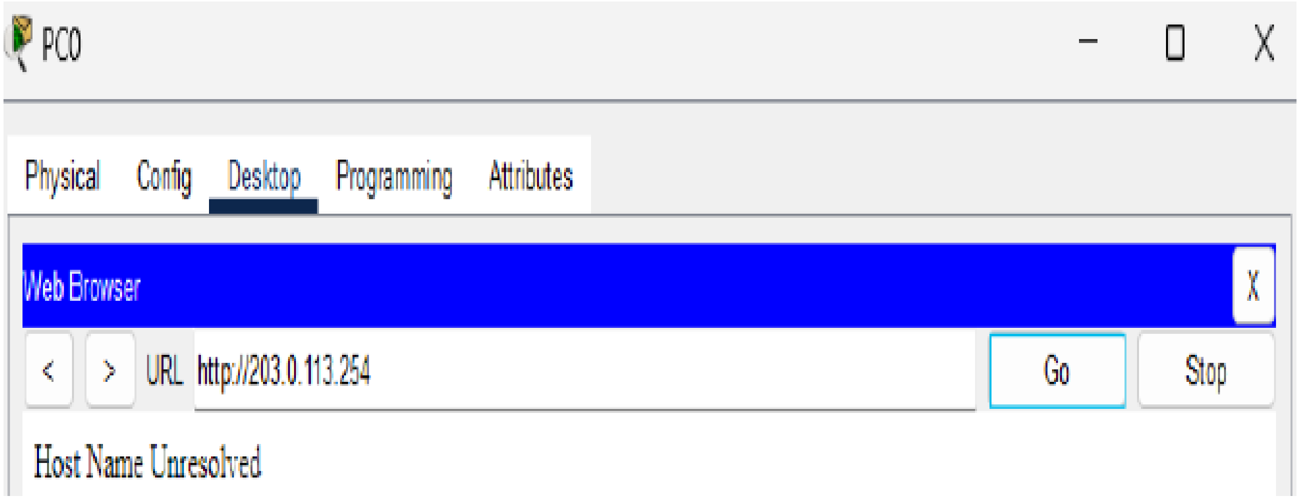
6. Final HTTP-Only ACL

- After ICMP testing, the ACL was replaced to allow only HTTP from the LAN to the outside:
  access-list INSIDE-ONLY-HTTP extended permit tcp 192.168.1.0 255.255.255.0 any eq www
  access-list INSIDE-ONLY-HTTP extended deny ip any any
  access-group INSIDE-ONLY-HTTP in interface inside

  This ensured that HTTP traffic was permitted, while ICMP and all other protocols were blocked.

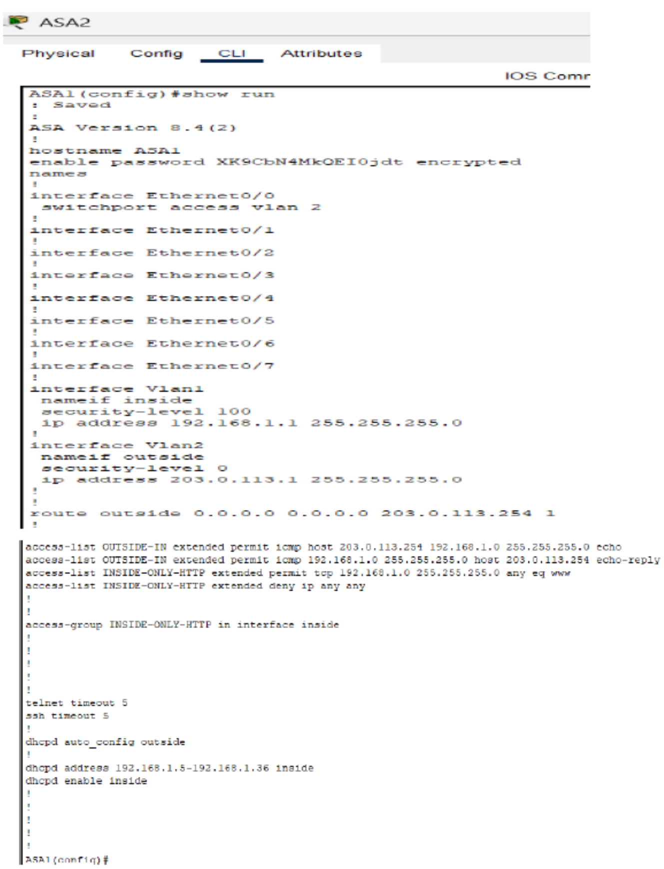


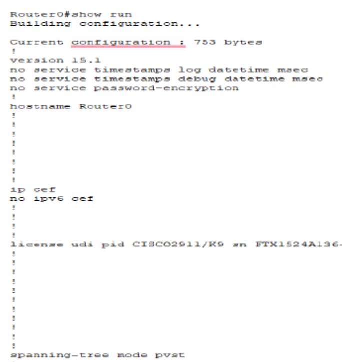


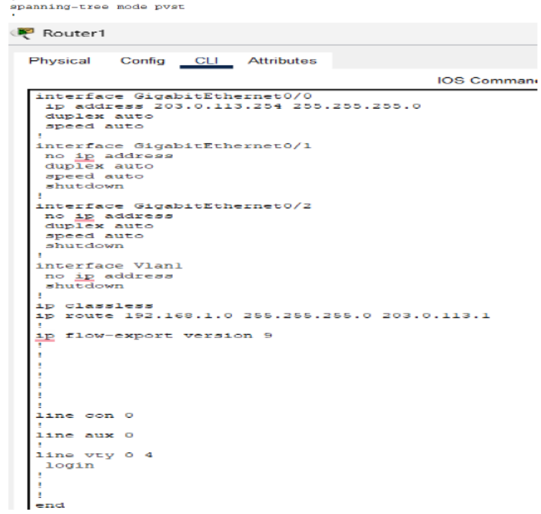


7. Final Configurations

- **Verification commands included:**
  - ASA: show run → confirmed hostname, interfaces, routes, and ACLs.
  - Router: show run → confirmed external IP and static route configuration.







8. Conclusion

The project successfully demonstrated how an ASA firewall enforces access control using ACLs. Initially, ICMP ACLs validated connectivity. Later, the final ACL restricted traffic to HTTP-only, showing that the ASA can effectively filter and manage network traffic. The objectives of securing the LAN and controlling permitted protocols were fully achieved.

9. References

- Cisco Networking Academy. (2023). Introduction to Network Security. Cisco Systems.
- Odom, W. (2020). CCNA 200-301 Official Cert Guide, Volume 2. Cisco Press.
- Cisco Systems. (2022). Cisco ASA Series General Operations CLI Configuration Guide.

  Retrieved from: https://www.cisco.com

  Packet Tracer :-
  https://drive.google.com/drive/folders/1Xeq0x8x9u20L0NQmViA_UJLXAPYcCp6u?usp=drive_link

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